

A Decomposition of Inflationary Financing

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July 2026

Abstract

Standard New Keynesian models are cashless: inflation finances deficits only through the erosion of nominal debt, and seigniorage plays no role. We relax this assumption by introducing money demand into a New Keynesian model under fiscal dominance. The resulting framework decomposes deficit financing into debt dilution, money creation, and the inflation tax. We find that seigniorage is a major source of fiscal financing, often more important than debt dilution, and that the composition of public liabilities significantly shapes the responses of inflation, interest rates, and output to fiscal shocks.

1 Introduction

Recent inflationary episodes throughout the developed world have revived the debate over the fiscal roots of inflation. One of the main thrusts of attack to this question has been the literature on the fiscal theory of price level (FTPL) as it manifests itself on the standard New Keynesian model. This paradigm features a cashless economy, where the only role of money is that of *numéraire*. Agents hold no money balances, and the exchange, savings and liquidity motives are absent.

This is an inherited feature of the models developed by Clarida et al. (1999), Woodford (2003) and others, where the economy is taken to the “cashless limit”. Because of this, the role of money in the economy is greatly diminished. In particular, these models feature no seigniorage: Inflation finances the fiscal authority only insofar as it inflates away nominal debt (the “denominator effect”). Thus, what was considered a central aspect of inflation in older traditions both Keynesian (such as Cagan (1956)) and Monetarist (Sargent and Wallace (1981)) is overlooked in the modern debate.

This note studies fiscal financing in a New Keynesian model with money in utility. Fiscal policy is active and monetary policy passive, so deficits are not fully backed by future sur-

pluses. Introducing money demand reopens channels of seigniorage that are absent in the standard cashless FTPL framework.

We then study how an unpaid for deficit is financed. Under monetary dominance, or when money is reduced to a unit of account, is trivial: surpluses offset deficits in the first case, and the price level raise so as to offset the newly acquired public debt and its interest bill in the second.

In our setup governments enjoy a new source of financing. Money demand by consumers allows for seigniorage to be raised via both direct money creation and the inflationary tax on existing money balances. Inflationary financing is no longer a single object, but clearly decomposable into three different components, all of them quantitatively relevant not only to achieving debt stability but also to the movements of output and inflation after the initial deficit episode.

We characterize how debt financing varies with monetary-policy responsiveness and with the steady-state composition of public liabilities. As figure 1 shows, under a steady state debt-to-money ratio of 1.2 the role of money creation increases rapidly as the monetary authority becomes more hawkish, and then peaks and decreases near the boundary $\phi_\pi = 1.0$. Both the debt dilution and inflation tax on money balances contributions to financing decline at first, and recover near the boundary.

Debt dilution – which is the only channel present in the usual formulations of FTPL models – is seldom responsible for more than half of the total financing, meaning it may be a mistake to trivialize the other forces. This note argues in favor of taking a broader view of the role of money and the possible channels of public financing in macro models.

2 Model Overview and Calibration

We work with a standard New Keynesian model, with the one difference of a money-motive being added to the agent’s utility function. It is taken entirely from Galí (2008). The key features are, as usual, that firms are in Dixit-Stiglitz monopolistic competition, prices are updated under a Calvo process, labor markets are perfectly competitive and the monetary authority follows a policy rule. This rule will not adhere to the Taylor principle. Lastly, the fiscal authority does not stabilize the debt; it implements a balanced budget subject to shocks. This will mean shocks that lead to deficits are not financed by future surpluses, and must be accommodated by monetary policy.

We will explore variability in three choice parameters: the degree of responsiveness of monetary policy to inflation ϕ_π , and the steady state levels of public debt and money balances, $\bar{b}$ and $\bar{\mu}$ respectively, both as percentages of output.

Figure 1

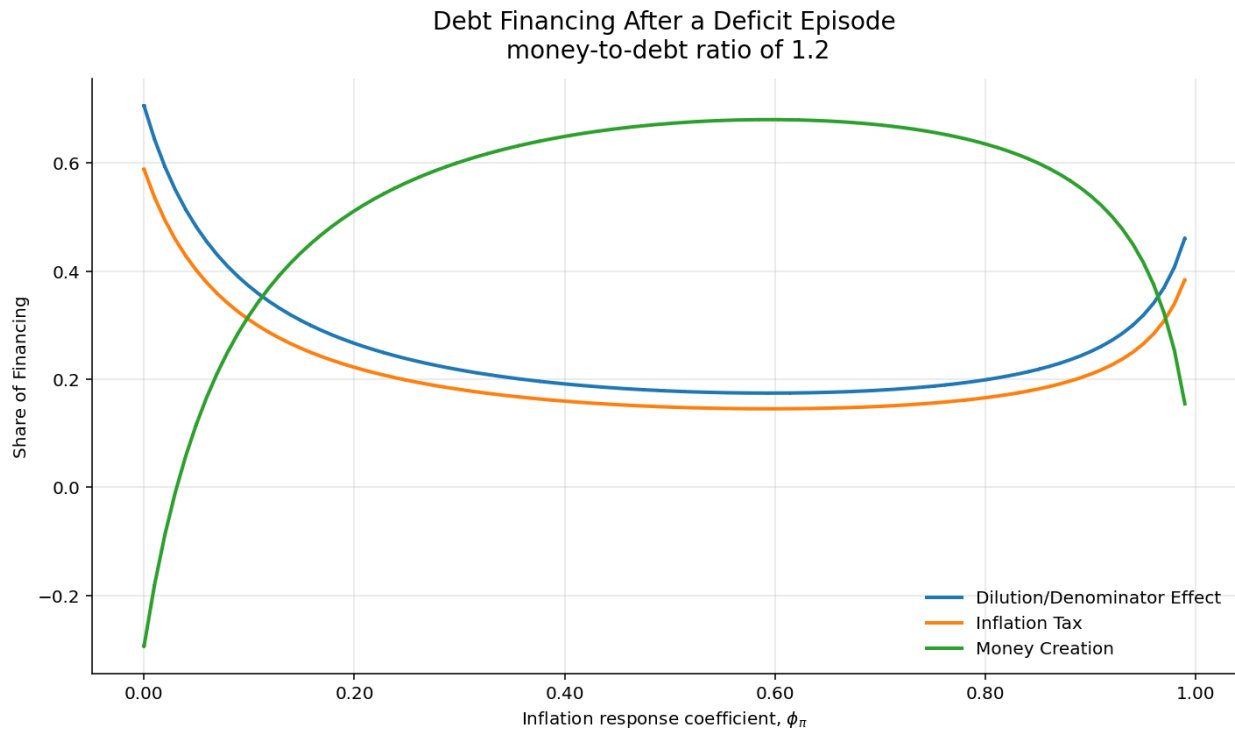


Table 1 gives summary statistics for public debt and money balances as shares of GDP for 12 countries from 1995 through 2022, with Eurozone countries being dropped from the panel upon entry in the Euro area. Broad money is M2 or M3, according to data availability. These will be our references for which values to explore when sweeping through the parameter space for $\bar{\mu}$ and $\bar{b}$.

Table 1: Summary Statistics for Money and Public Debt (% of GDP)

Statistic	M1	Broad Money	Public Debt
Count	254	254	254
Mean	35.9	97.4	71.3
Std. Dev.	31.6	56.7	51.1
Minimum	4.4	19.9	7.8
Median	27.3	77.1	61.0
Maximum	182.9	278.0	258.4

The ratio of public debt to money balances plays a key role in determining how the model responds to fiscal shocks. Figure 2 displays this ratio for the full panel, giving us a notion of the relevant parameter space on which to focus.

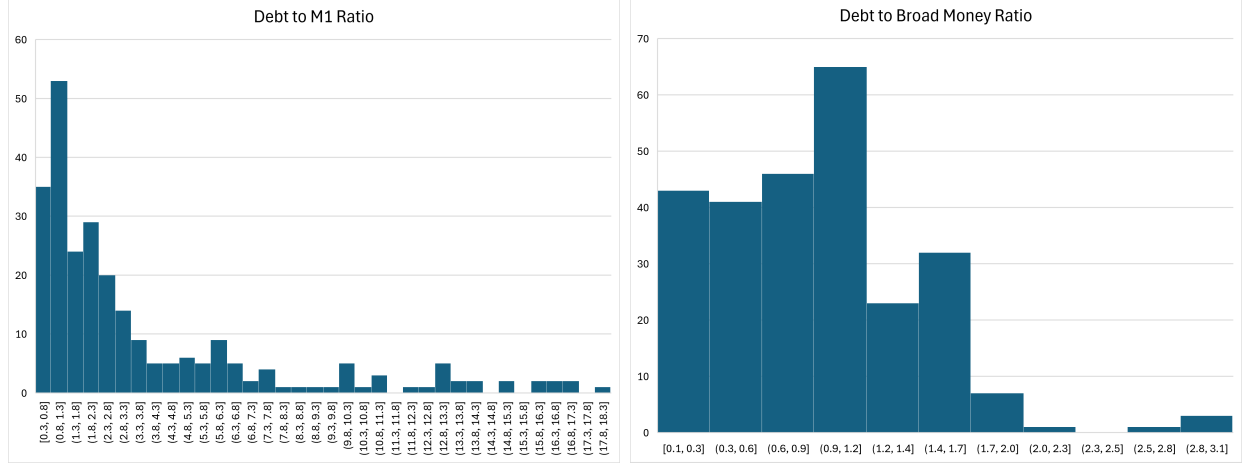


Figure 2

As for ϕ_π , we will explore the full $[0, 1)$ space in all cases. With these disciplining observations as groundings for our parameter exploration we can proceed to results.

3 Results

Results are obtained by solving the model for each $(\phi_\pi, \bar{\mu}, \bar{b})$ triple, chosen from gridded values spanning the relevant parameter space and obtaining the fiscal shock IRF for each of the relevant outcome variable.

Outcomes are clustered in two groups: “macro”, which encompass output, inflation and interest rates; and “debt financing”, the shares of public debt contracted after the shock financed via debt dilution, money creation, and the inflation tax.

3.1 Macro block

Figure 3 displays the total accumulated gap in each macro variable resulting from an initial deficit. We compare four different sets of steady state money and debt: A “low-low” setup figures both at 0.5 of output; a “high-low” figures $\bar{\mu} = 1.0$ and $\bar{b} = 0.5$; a “high-high” has both at 1.0; and a “low-high” figures $\bar{\mu} = 0.5$ and $\bar{b} = 1.0$.

In all cases all variables increase as ϕ_π increases, up to a limit at some value near 1.0, after which they all begin declining precipitously. Inflation and interest peak at nearly the same value, consistent with Fisherian dynamics under FTPL. The output gap response is mostly increasing in ϕ_π , and peaks more quickly than the other two.

Contrasting the “high-high” scenario to the “low-low” one, it is clear that the overall shape and relative magnitudes of the shock on interest rates, inflation and output gap are

invariant between the two. Holding the debt-to-money ratio constant leaves the shape of impulse responses nearly unchanged.

Absolute magnitudes roughly double when going from the “high-high” to the “low-low” scenario, as public liabilities have a cushioning effect in this model. Under high debt, a little inflation generates a large dilution in overall debt; under high money balances, a little inflation tax generates enough revenue to offset the initial deficit. In this sense, debt and money are taxable bases, and the larger they are the lower the tax rate – inflation and interest rates — the monetary authority has to impose to finance the fiscal shock. The intermediary setups illustrate how changing the ratio of money to debt can affect the overall macro response.

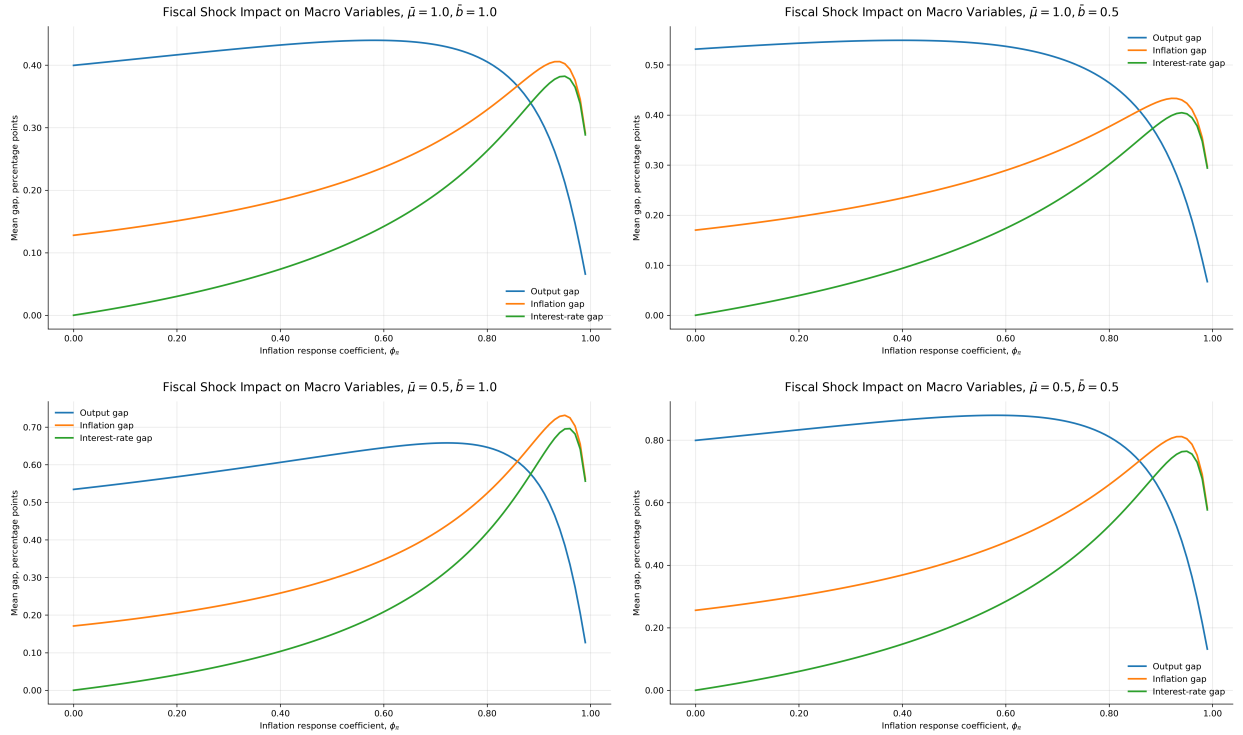


Figure 3: Macroeconomic outcomes under alternative debt and money calibrations.

3.2 Debt Financing

Figure 4 plots heatmaps for the share of debt that is financed by each of the inflation tax on money balances, direct money creation, and debt dilution. The Y axis tracks the debt-to-money ratio, while the X axis tracks the degree of inflation responsiveness by the monetary authority.

It spans the relevant parameter space, and can be considered a generalization of figure 1, which showcased only the 1.2-ratio scenario.

Although magnitude is relevant for the overall impact of the shock, the debt-to-money *ratio* is a sufficient statistic to characterize financing shares: an economy that has high levels of both money and debt, but in which debt is 2 times as prevalent will finance itself in the same way as one with low levels of both but the same 2 to 1 ratio.

The first image in figure 4 plots the role of money creation, which displays a hump-shaped response to ϕ_π . It is initially negative, becomes the dominant financing source over much of the parameter space, and declines as $\phi_\pi \rightarrow 1$. Because it does not operate through existing liability stocks, it is comparatively insensitive to the debt-to-money ratio.

Although it is completely absent from standard formulations of the model, money creation is the single most important channel of financing for much of the parameter space.

The inflation tax figure is dominated by the top-left region of the plot, which combines very low levels of debt relative to money with money destruction following the fiscal shock. In these cases, the taxable base for the inflation tax is large and there is little money creation, so most of the financing happens through it. From there its role declines quickly, only rising again on the mid-to-top right of the plot, where a high tax base combines with higher interest rates (and thus inflation) and leads to the inflation tax once again playing a more important part.

The overall share of the inflation tax – which is also absent from standard formulations of the model – never falls much below 20% of the total financing of debt.

The debt dilution channel is the most dynamic one. In it we see how for all levels of ϕ_π an increase in the debt-to-money ratio leads to higher dilution financing; how this channel is most significant for very high and very low values of ϕ_π , losing importance in the middle sections where money creation dominates; and how it almost vanishes when money balances are sufficiently dominant in the economy.

Crucially, it is only in very narrow bands of the parameter space – the bottom left and bottom right ones – that debt dilution plays an outside role in financing. It is otherwise an auxiliary force in restoring debt stability.

4 Conclusion

Debt dilution is the sole mechanism through which inflation finances fiscal imbalances in standard FTPL models. Introducing money demand changes this dynamic, as seigniorage through money creation and inflation taxation becomes quantitatively important and frequently dominates debt dilution. Output, inflation, and interest rates become sensitive to the aggregate public sector balance sheet, which has a cushioning effect on the economy. Given the empirically high levels of money balances in modern economies, the additional

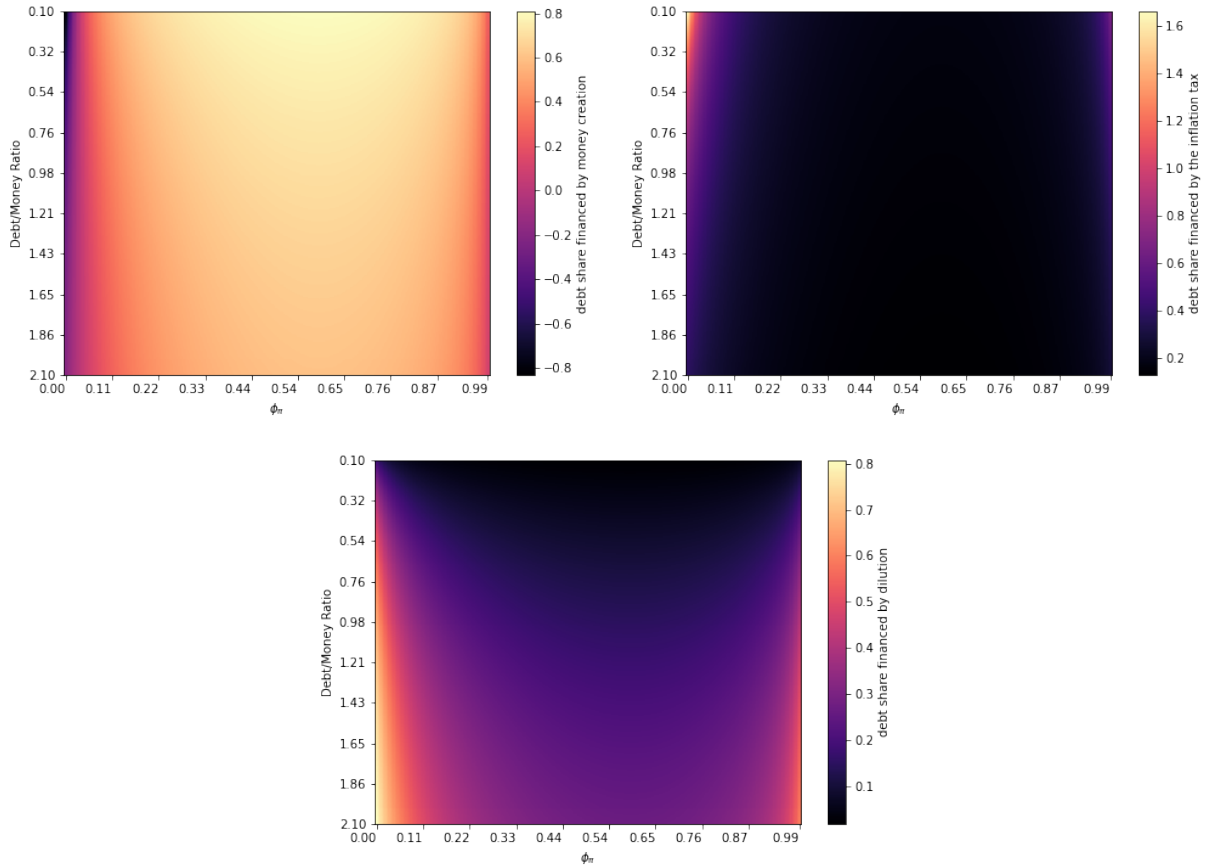


Figure 4: Decomposition of Debt Financing After a Fiscal Shock

cushioning provided by these balances is also quantitatively relevant. These findings suggest that cashless FTPL models may substantially understate the role of money in fiscally dominated regimes.

This is not novel. The pre New Keynesian literature was keenly aware of the role seigniorage played in financing governments, especially non stabilizing ones. That these insights have fallen out of perception is mostly an artifact of more recent modeling choices that have become mainstream in a period where monetary-fiscal interactions were perceived as less important. As this ceases to be the case, we expect to see more works highlighting the role of money in public finances and how it interacts with the overall economy in short-run macro models.

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